TOTAL CHAOS

24621

Engineering Portfolio 2025 – 2026

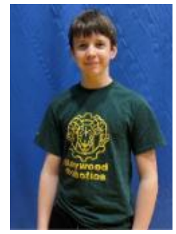
FTC 24621
Maywood Middle School
Renton, WA.

Team Members

Aman Jacob
Dismantled old robot. Worked on portfolio year-round.

Darren Zhou
Drive Coach
Secondary Operator
Human Player, Hardware

Gordon Mathis
Chassis Assembly
Human player



Liam Krell
Robot construction, CAD, Primary Driver

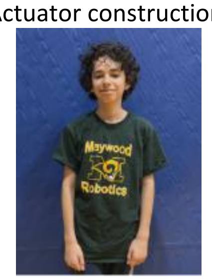
Oslo Griebling
Prototype and design, Helpful with many different tasks.

Romeo Duncan
Created color picker for RGB system with AI. Actuator construction.

Ethan Vital
Assembly and Disassembly.

Peter Hutton
Disassembly of last year's robot

Vihaan Srinivasan
Driver and Autonomous Code



Nathaniel Woodcock
Hardware, helpful with many different tasks.

Spencer Levy
Chassis Construction

Abby Soliven and Evren Spradling
Helped with Portfolio and Tri-Fold Board

About Our Team

24621 Total Chaos is one of two FTC teams from Maywood Middle School in Renton, Washington, along with our sister team, 24620 Deep Charge. Most of our members have prior FTC experience, allowing us to work efficiently and build on past seasons. This is Maywood's third year participating in FIRST Tech Challenge, beginning with our rookie season in 2023.

Our Mentor!

Ted Griebling: He is a mechanical engineer in the robotics field who served as a mentor for our FTC team since last season. He attended Lafayette college, and got a degree in mechanical engineering. He also has prior experience volunteering in FRC. He guided our team in mechanical design and engineering.

Through his guidance, we improved our robot design, learned hands-on troubleshooting strategies, and gained insight into real-world engineering practices. These connections have strengthened our technical skills, exposed us to the broader engineering community, and inspired us to share our knowledge with others, applying our technological skills to benefit both our team and our local community.

Engineering Goal

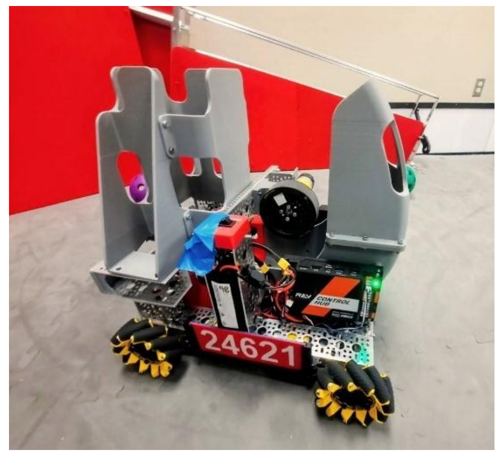
Design a **fast and reliable robot with accurate** scoring in Autonomous, TeleOp, and Endgame. Our priorities were **consistency, driver control, and mechanical reliability** while staying within our constraints.

Robot Design Summary

Our robot features a **compact 16x16 Mecanum drivetrain** paired with a **weighted single-flywheel shooter and roller intake system**. This enables fast all-around movement, efficient artifact collection, and accurate scoring from multiple positions on the field.

Key Features of Major Subsystems

Drivetrain



- Block coded field-centric control.
- Precision and turbo speed modes for driver flexibility

Intake & Storage

- Laser distance sensor cuts intake power after 3rd artifact
- Vertical magazine and servo paddle to minimize jams

Scoring Mechanism

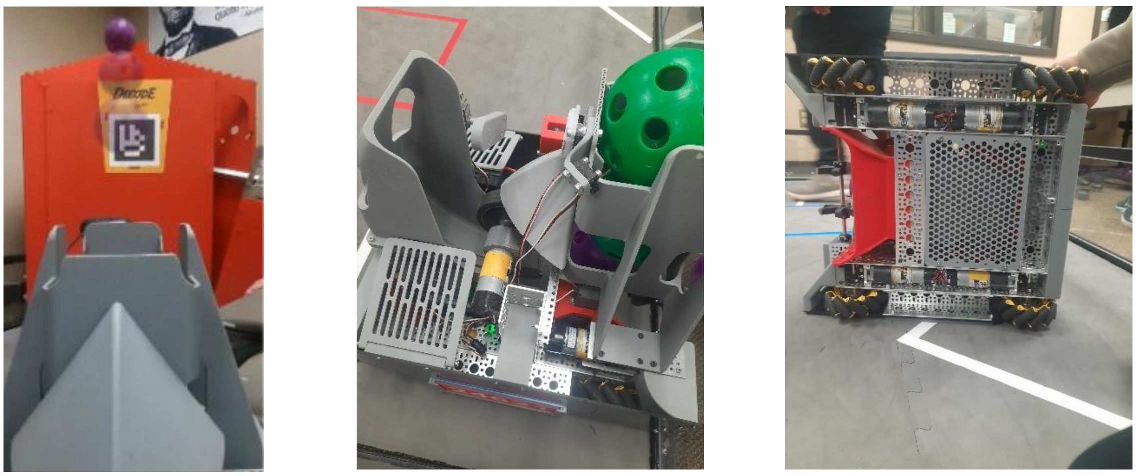
- Weighted single-flywheel shooter chosen after evaluating multiple designs
- PID Tuned for consistent velocity and repeatable shots
- RGB indicator reports when April tag is visible, and flywheel is at optimal scoring velocity.

Autonomous & Vision

- Autonomous routine capable of scoring up to 9 artifacts
- April Tag-based auto-aiming and flywheel control for improved scoring accuracy from both firing lines.

Engineering Benefits

1. **Weighted Single Flywheel** – Conserves momentum and provides the best balance of speed, accuracy, reliability, and cost compared to double flywheels, catapults, and regular unweighted flywheels.
2. **Compact Chassis** improves maneuverability and simplifies subsystem integration reducing chance of failure.
3. **Software-Assisted Control** – Auto-aiming, visual feedback, and field-centric driving reduce driver workload, and increase scoring consistency.



Our deflection systems and compact chassis

Iteration & Improvement

Early designs experienced inconsistent artifact handling which caused penalties during competition. Through prototyping, testing, and redesign, we improved intake reliability, shortened cycle times, and increased scoring consistency. Each revision was guided by performance testing and match feedback.

Game & Competition Performance:

Auton Potential: 42 Points 9 artifacts scored	TeleOp Potential: 81 Points 27 artifacts scored
Endgame Potential: 20 Points	Total: 143 Points without alliance

Engineering Takeaway

We prioritized simple, reliable solutions that could be refined and trusted under match conditions. Our iteration, testing, and match analysis resulted in a design that performs consistently in competition. Our robot has evolved into a cohesive system that reflects strong engineering fundamentals and the values of FIRST.

Mentoring for a Sustainable Future

This year, we have been able to expand our program from 20 students to 30 and include a learning cohort of sixth grade students. Eighth grade students come in on a separate day to mentor this cohort and help them build and program their own robot. Next year they will be ready to join the competition team as our eighth-grade students transition to high school.

Eight grade student Liam K. helps 6th grade student Kingston H. design and build a robot (left)
6th graders Samika S. Christina G. and Nandana S. test encoder drive functions (right)

Outreach

Our team does several activities to raise interest in robotics in our community. For example, we went to Apollo Elementary and Maple Hills Elementary STEM fair nights to foster curiosity for robotics to approximately 50 – 100 younger students and informed them about FTC and how to join it when they reach middle school. We also attended Sea Fair where people of all ages got a chance to drive our robot in the Boeing tent.

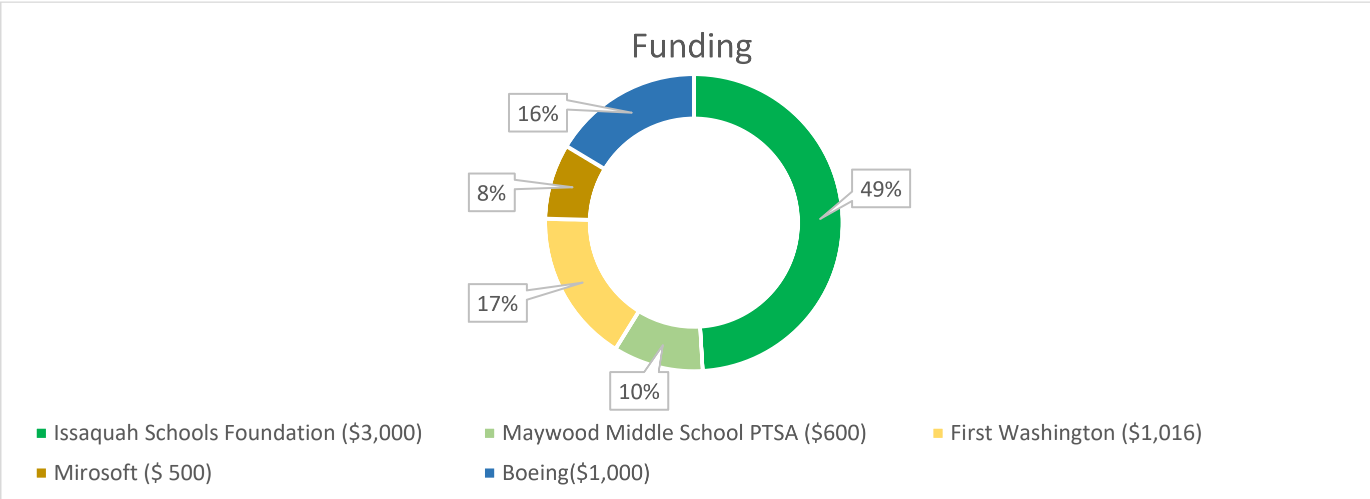
Recruitments to our program this year include younger students from previously visited elementary schools, Lego robotics students, and **mentors recruited for our sister team, 24620 Deep Charge**. Our team plans to visit more elementary schools such as Newcastle or Briarwood, to further reach out to others, and inspire people of all ages in the STEM community.

Sustainable Funding

After OSPI cut funding for robotics, we asked for and received increased program funding from our PTSA who purchased the game kit for us which has allowed us to meet this year in our own building instead of a nearby high school.

When Issaquah Schools Foundation initially wanted to cut funding for robotics by $250 per school this year, we were able to **successfully convince** them to maintain giving at current levels **for all schools in our district**. We have also received increased giving through Benevity matching dollars from Microsoft this year as well as grants from Boeing and First Washington.

We are grateful for our sponsors' support. Our program is entirely grant funded. Without them, we would not be here



A few members of Maywood Robotics 24621 Total Chaos and 24620 Deep Charge show gracious professionalism for support from all the sponsors.

Gracious Professionalism

We believe that success isn’t measured only by engineering performance, but also by how we treat others. We strive to show respect, honesty, and kindness—whether we are collaborating within our team, mentoring younger students, helping another team troubleshoot during a match or helping our sister team with anything. Gracious Professionalism means celebrating the successes of others, learning from challenges with a growth mindset, and sharing knowledge without expectation of reward. For us, it’s about building a supportive community where everyone can grow, on and off the field.

Reflecting on Past Experience

Over this season and the last two years, our team has grown and improved both our teamwork and technical approach. We found that a smaller, dedicated drive team allows for more effective practice and stronger competition performance. We also learned that high team engagement directly drives progress, strengthening both our robot and our team culture.

We stay organized easily by dividing roles among our team based on experience and skills. However, we also make sure that everyone has a chance to learn and expand their knowledge, as well as develop leadership skills.

Additionally, early competition experience highlighted the value of a strong autonomous program, inspiring us to invest more time in innovation and refinement.

Future Goals

1. Strengthen our problem-solving abilities by identifying challenges, analyzing their causes, and collaboratively developing effective solutions.
2. Develop proficiency with Android Studio to support efficient robot programming and debugging.
3. Build on lessons learned from previous seasons to significantly improve our performance and compete with a robot capable of placing in the top five.

Tracking Progress

We track our progress by evaluating how effectively we meet and exceed our established goals. For example, we assess our ability to consistently score game elements by implementing high-performance strategies developed through testing and iteration (trying over and over again).

Learnings

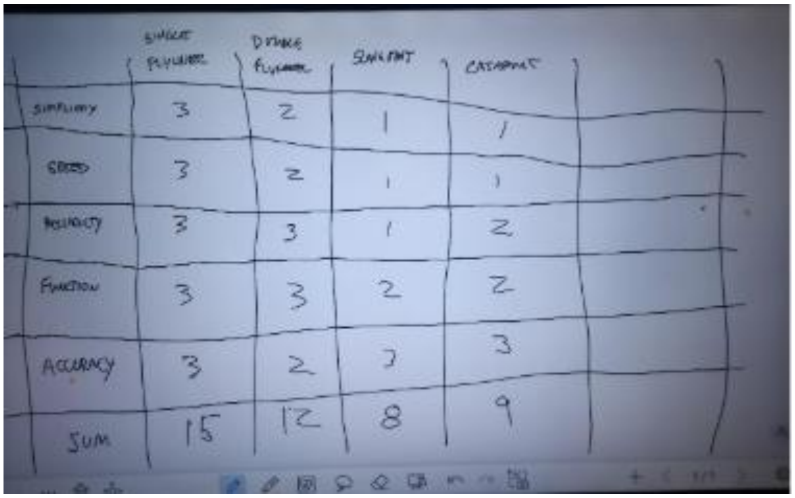
Our mentor makes sure we are engaged in every practice, and that we show interest and have fun in FTC robotics. A life lesson we have learned is that you should never expect something to work on the first try, especially with software. Patience is possibly the best quality for a team member.

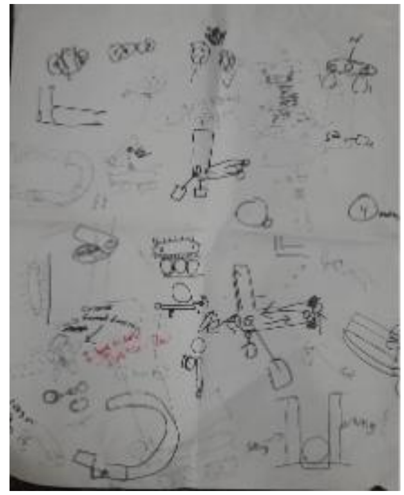
Our mentor, Ted Griebling, explains software to Liam K and Romeo D.

Design Process

At first, nobody agreed on a specific design; we had many different ideas ranging from rubber band intakes, Archimedes screws, or oversized catapults. Many ideas were scrapped due to limited resources, design constraints, expense, or inefficiency. We used several decision making matrices to help us come up with ideas.

Mentor Ted G. helps Ethan V. with the intake assembly and motor on the back of the robot.

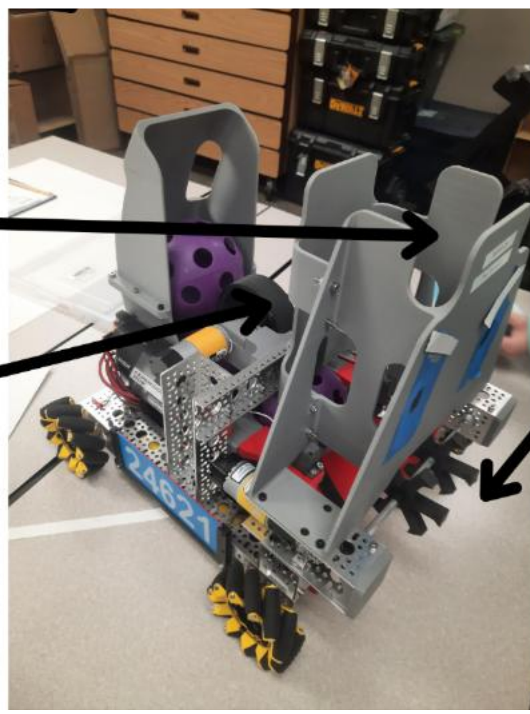




An example of our decision matrices

Storage
How our storage works is that after the ball took in by the roller it is pushed up. As it is pushed up the ball is stuck spinning in the storage. The storage can hold up to 3 balls, and when it is ready to release the roller goes in reverse and it is shot.

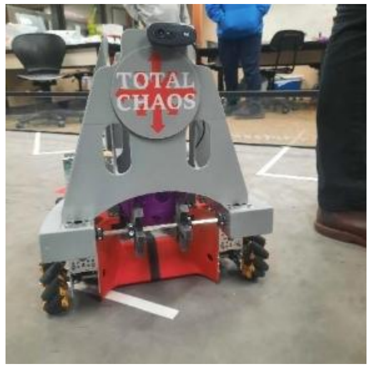
Launching
For launching I have a 2678-3857 RPM motor that spins from above to speed up the ball. Then onto a curved ramp to launch it.



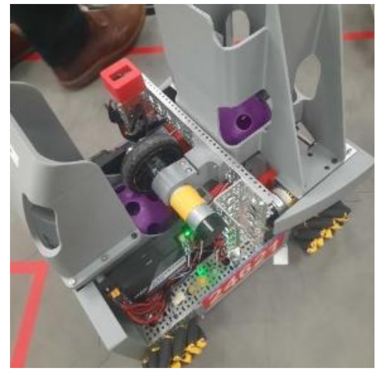
Intake
For intake we have a roller that turns clockwise to suck the balls in. it goes slowly counterclockwise
 with two other wheels to to get it ready to shoot.

Our Robot (This has changed over the course of many weeks)

Side View



Front View


Interior View

Problem Solving

Our team follows a structured, iterative design process to solve problems. We identify root causes and use design matrices to evaluate tradeoffs between performance, reliability, and complexity. When problems could not be fully eliminated, we designed subsystems to minimize their impact on overall robot function.

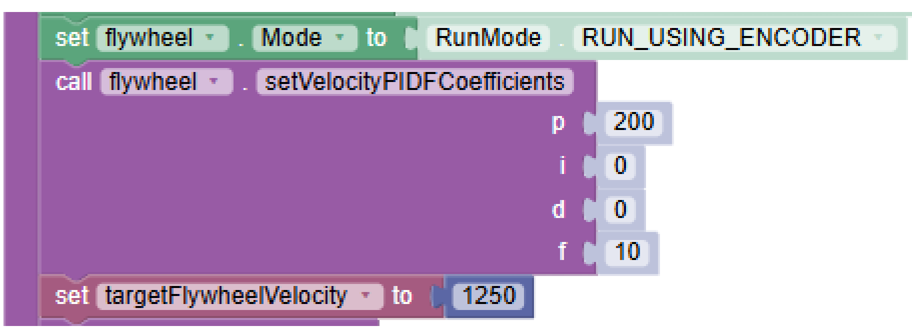
For example, we had artifacts going into our robot causing us to get penalties during competition, below you can see the transformation of the artifact dislodging system to avoid the penalties. We improved our design by making a makeshift blocker and then improved to a 3D printed “blocker” that uses servos to lift it anytime we shoot so it doesn’t get in the way. We also added “ribs” that prevent the ball from getting lodged in the area for our components. Finally, we added a laser distance sensor at the top of our magazine to cut power to the intake before it grabs a 4th artifact.

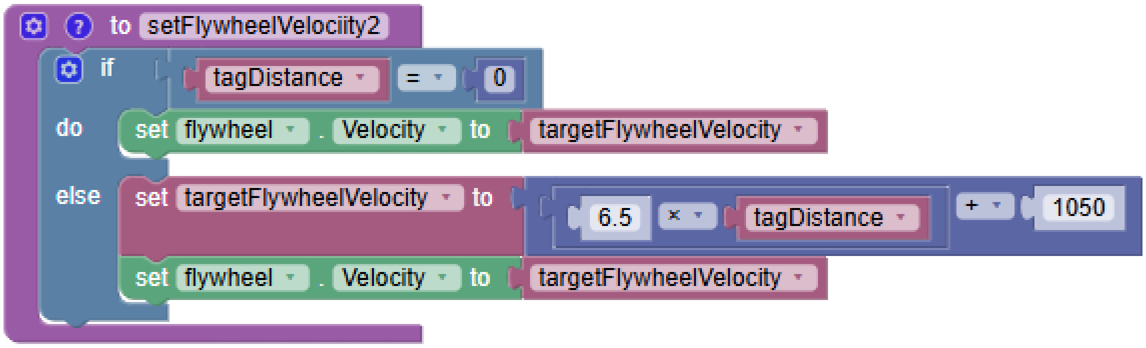
Before	>	After

Our approach to problems resulted in a robot capable of efficient artifact collection with minimal jams, accurate scoring from multiple field positions, and fast cycle times within size and weight limits. Compared to previous designs, our robot demonstrates improved consistency, reliability, and overall performance.

Autonomous Program and Auto aiming

Our autonomous scores 9 artifacts and leaves the shooting zone, scoring up to 42 points. We have also developed an auto aiming system to detect April tags whenever the robot can line up to shoot to get an accurate shot from any position. The shooting wheel gets ready and from anywhere on the field can accurately shoot into the targeted goal.





A portion of code that illustrates our algorithm for automatically adjusting flywheel velocity based on distance to the goal.

Photos from our season


Our drive team get ready for a match


The team brainstorms designs and creates a plan


Team members test robots in the pit


Victory at Interleague

Maywood Middle School Robotics | November 2, 2025, Team 24621 Total Chaos & Team 24620 Deep Charge